Beyond Filtered Scores: Does Benchmark Filtering Reduce Scores More Than Expected?

Research Gap Statement

Large language models (LLMs) are increasingly evaluated on automated program repair (APR), but benchmark scores may be affected by data leakage. Zhou et al. (2025, Table 2) identified potential overlap between 53 of 500 SWE-bench Verified instances (10.6%) and StarCoder's pre-training data. Wu et al. (2026, Table 6) reported relative resolution-rate decreases of 20.0%–28.6% for the top-10 SWE-bench Verified submissions on the 1–4 hour difficulty subset after filtering. These findings raise concerns about benchmark reliability and motivate research into leakage-aware evaluation (Zhang et al., 2026, Sec. 8.1).

However, score decreases after filtering may also result from changes in instance difficulty and benchmark composition. BenchChecker reports results by difficulty band, but no random or difficulty-matched removal control was found in the main text and supplementary sections reviewed. This leaves unresolved whether removing flagged instances decreases performance more than removing an equivalent number of unflagged instances with comparable difficulty. This study addresses that question specifically for the examined SWE-bench Verified flag set, without assuming that the flags prove model-specific memorization.

Existing methods identify different forms of potential exposure. BenchChecker uses repository-presence probing and patch-persistence checks with model-specific cutoff dates, whereas Zhou et al. (2025) examine overlap with StarCoder's pre-training data. Their flags are not directly comparable because they use different criteria and reference data. Neither method proves that a particular evaluated model memorized a flagged instance; likewise, a null result would not establish the absence of contamination. A difficulty-matched removal control can estimate whether score decreases exceed those expected from changes in benchmark composition, although it cannot establish causation.

This study proposes a secondary analysis of SWE-bench Verified, a 500-instance subset of SWE-bench (Jimenez et al., 2024). It will first verify whether Zhou et al.'s 53 candidate flags can be mapped to official instance IDs and whether complete per-instance resolution outcomes are available from public submissions (SWE-bench, n.d.). If feasible, it will compare the resolution-rate decrease after removing flagged instances with decreases from repeated removals of the same number of unflagged instances matched by difficulty. The 1–4 hour and >4 hour bands will be combined into a ≥1 hour group for matching because BenchChecker reports only three instances in the >4 hour band (Wu et al., 2026, Table 6). **Excess decrease** is the observed decrease minus the mean decrease across matched-removal samples. Results will include percentage-point and relative decreases with uncertainty estimates. If reliable flags or complete outcomes are unavailable, the design will be narrowed.

Research Questions

RQ1. Do flagged and unflagged instances have different resolution rates after accounting for difficulty?

- **Independent variable:** Exposure-screening status (flagged/unflagged).
- **Dependent variable:** Resolution rate (%).
- **Analysis:** Compare resolution rates between flagged and unflagged instances within difficulty groups, combining the two highest-duration bands.

RQ2. Does removing flagged instances produce a larger resolution-rate decrease than removing the same number of unflagged instances matched by difficulty?

- **Independent variable (IV):** Removal condition (flagged-instance removal or matched unflagged-instance removal).
- **Dependent variable (DV):** Excess resolution-rate decrease, measured in percentage points.
- Analysis: Calculate each decrease as the full-benchmark baseline resolution rate minus the resolution rate after removal. Compare the observed decrease after flagged-instance removal with the mean decrease across repeated matched-removal controls, using the same baseline for both. Report relative decreases and uncertainty estimates.
- **Hypothesis:** Flagged-instance removal produces a larger resolution-rate decrease than the matched control.

RQ3. Does excess resolution-rate decrease differ across difficulty groups?

- **Independent variable:** Difficulty group.
- **Dependent variable:** Excess resolution-rate decrease (percentage points).
- **Analysis:** Compare group-specific estimates and uncertainty, interpreting small groups cautiously.

References

Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. R. (2024). *SWE-bench: Can language models resolve real-world GitHub issues?* In *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=VTF8yNQM66

SWE-bench. (n.d.). *Experiments repository.* GitHub. https://github.com/swe-bench/experiments

Wu, D., He, X., Wang, S., & Sun, K. (2026). *BenchChecker: Assessing the credibility of bug-fixing benchmarks for LLMs.* In *Proceedings of the USENIX Security Symposium 2026* (pp. 5077–5096). https://arxiv.org/abs/2610.19447

Zhang, Q., Fang, C., Xie, Y., Ma, Y., Sun, W., Yang, Y., & Chen, Z. (2026). *A systematic literature review on large language models for automated program repair* [Preprint]. arXiv. https://arxiv.org/abs/2405.01466

Zhou, X., Weyssow, M., Widyasari, R., Zhang, T., He, J., Lyu, Y., Chang, J., Zhang, B., Huang, D., & Lo, D. (2025). *LessLeak-Bench: A first investigation of data leakage in LLMs across 83 software engineering benchmarks.* arXiv. https://arxiv.org/abs/2502.06215